

Parkinson's Disease Drug Repurposing — Deep Validation (Retained Candidates)

1. OLTIPRAZ

1.1 Candidate identity

Oltipraz — CHEBI:77319 (confirmed via Monarch); CAS 64224-21-1; synonym RP-35,972; a synthetic 1,2-dithiole-3-thione. Historically an FDA-investigated chemopreventive/antisteatotic agent (liver disease, lung cancer prevention). No history of PD-directed development.

1.2 Proposed mechanistic hypothesis

Oltipraz activates Nrf2 (NFE2L2), which induces NQO1/HO-1 expression and the canonical anti-ferroptotic module (GPX4 $\uparrow$, SLC7A11 $\uparrow$, ACSL4 $\downarrow$) in CNS tissue, reducing oxidative stress/ferroptosis and suppressing microglial activation. This CNS pharmacology is oltipraz-specific but was demonstrated only in acute-injury paradigms (a nerve-injury/neuropathic-pain model and a cerebral ischemia-reperfusion model) — never in a PD, MPTP, 6-OHDA, or α -synuclein model. The link to PD is an inferred bridge: the same Nrf2 $\rightarrow$ GPX4/NQO1 axis and anti-microglial effect are independently protective for dopaminergic neurons in actual PD models, but there driven by different compounds (melatonin, schisanhenol, β -caryophyllene).

This revises the originally-proposed initial mechanism. The initial framing — oltipraz acting through a "parallel non-mitochondrial FSP1–CoQ10 axis" — has **no oltipraz-specific evidentiary support** in PubMed at all. The real, verifiable oltipraz-CNS pharmacology is

GPX4/SLC7A11/ACSL4-centric (the canonical ferroptosis-defense axis), not FSP1/CoQ10-centric. This is a genuine mechanism revision, not a refinement.

1.3 Complete mechanism chain

1. Oltipraz → activates Nrf2 (NFE2L2)
2. Nrf2 → induces NQO1/HO-1 in CNS tissue (spinal cord, hippocampus, prefrontal cortex)
3. Nrf2 → upregulates GPX4/SLC7A11, downregulates ACSL4 (antiferroptotic program), in brain tissue (MCAO/R model) and PC12 cells
4. Oltipraz (via this axis) → suppresses microglial activation (CD11b/c↓) in CNS tissue
5. This Nrf2-driven axis → reduces oxidative stress / ferroptosis in neural tissue (oltipraz-specific, non-PD models)
6. **[Inference]** The same Nrf2→GPX4/NQO1 axis and anti-microglial effect, independently shown (other compounds) to protect dopaminergic neurons in PD/MPTP/α-synuclein models → hypothesized to translate to oltipraz's own PD-relevant neuroprotection — **never directly tested**

Secondary, weaker corroborating strand (target-class, not oltipraz-specific): NQO1 interacts with PRKN (Parkin, a bona fide causal PD gene per Monarch's curated OMIM/Orphanet associations) and AKT1; Akt-mediated phosphorylation of NQO1 triggers Parkin-mediated degradation, reducing NQO1's antioxidant capacity as PD progresses.

1.4 Edge-Level Evidence Ledger

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
1	Oltipraz	activates	Nrf2 (NFE2L2)	OBSERVED_LITERATURE	PubMed	PMID 37535075	ABSTRACT	Oltipraz described as an Nrf2 activator inducing NQO1/GCL, protecting mouse liver

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								from acetaminophen injury via GSH recovery (non-CNS confirmation of core pharmacology).
2	Oltipraz	induces (Nrf2-dependent), CNS tissue	NQO1, HO-1 (spinal cord, hippocampus, PFC)	OBSERVED_LITERATURE	PubMed	PMID 31234342	ABSTRACT	Mouse sciatic-nerve-injury (neuropathic pain) model: systemic oltipraz enhanced Nrf2/HO-1/NQO1 in spinal cord, hippocampus, PFC — confirm

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								s oltipraz reaches and engages this pathway in CNS tissue. Also decreased p-Akt/PI3K signaling. Not a PD model.
3	Oltipraz	upregulates GPX4/SLC7A11, downregulates ACSL4, Nrf2-dependently	antiferroptotic program (brain, MCAO/R; PC12 cells)	OBSERVED_LITERATURE	PubMed	PMID 39096875	ABSTRACT	Oltipraz reduced infarct volume/oxidative stress/ferroptosis in mouse cerebral ischemia-reperfusion; Nrf2-KO abolished the effect

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								(causal) · Canonical GPX4-centric axis, not FSP1/CoQ10. Not a PD model.
4	Oltipraz	suppresses	microglial activation (CD11b/c), CNS tissue	OBSERVED_LITERATURE	PubMed	PMID 31234342	ABSTRACT	Same neuropathic-pain study: oltipraz inhibited microglial activation and PI3K/Akt, NF-κB, MAPK overexpression in spinal cord. Not a PD model.
5	GPX4-centric antiferro	protects	dopaminergic neurons	OBSERVED_LITERATURE	PubMed	PMID 38488331	ABSTRACT	Different compound

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
	ptotic activity (Nrf2-driven, other compounds)		, PD/MPTP models	URE (class-level, NOT oltipraz)		(melatonin, Sirt1/Nrf2/HO-1/GPX4, α -syn PFF model — NQO1 not tested); PMID 41460594 (schishenol, Nrf2/TrxR1/GPX4, MPTP model — SLC7A11 not involved)		nds; NQO1/SLC7A11 combination oltipraz engages is not reproduced identically in either bridge paper — "same axis" framing overstates mechanistic overlap (critical finding).
6	NQO1 activity (other compounds / endogenous)	supports	dopaminergic neuron survival, PD models	OBSERVED_LITERATURE (class-level, NOT oltipraz)	PubMed	PMID 33271195 (β -caryophyllene, MPTP model); PMID 313586	ABSTRACT	Neither paper involves oltipraz. PMID 31358653's stage-dependent

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
						53 (NQO1 neuroprotective in MPTP + α -syn-transgenic mice; Akt phosphorylation $\rightarrow$ Parkin-mediated degradation abolishes protection; NQO1 protein "most prominent in active/early-stage PD, virtually absent at end-stage")		nt NQO1 finding raises a therapeutic-window concern independent of any delivery question.

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
7	NQO1 (HGNC: 2874)	interacts_with	PRKN (HGNC: 8607), AKT1	OBSERVED_KG — but not independent (see Critic finding)	Monarch	HGNC: 2874 ↔ HGNC: 8607 / AKT1, infores: biogrid	MONARCH_KG	BioGRID interaction edges. On inspection, the underlying supporting publications are PMID 31358653 (already cited directly) plus two PD-irrelevant cancer-biology papers (PMID 26540344 breast cancer/p53; PMID 31842909 hepatocellular

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								carcinoma/SIRT6) — this KG strand is largely circular re-indexing of the same primary evidence, not a genuinely independent corroboration.
8	PRKN (HGNC: 8607)	causes / gene_associated_with	Autosomal recessive juvenile PD 2 (MONDO:0010820); Young-onset PD (MONDO:0017279)	OBSERVED_KG	Monarch	HGNC: 8607↔ MONDO:0010820 (OMIM, causal), ↔MONDO:0017279 (Orphanet)	MONARCH_KG	Confirms PRKN is a genuinely well-annotated, bona fide causal PD gene (36 phenotype

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								associations including parkinsonism, Lewy bodies, SN gliosis, bradykinesia).
9	Oltipraz	(no relationship found in literature to)	PD / dopaminergic neurons / MPTP / 6-OHDA / SN / α -synuclein	UNRESOLVED (confirmed absence, twice-independently reproduced)	PubMed	zero-hit search, reconfirmed by both the reconstruction agent and the critic	ABSTRACT (search listing)	No compound-specific PD evidence exists anywhere in PubMed for oltipraz.
10	NQO1 C609T polymorphism (rs1800566)	NOT significantly associated with	Parkinson's disease	CONTRADICTED (corrects a prior mischaracterization)	PubMed	PMID 25054016	ABSTRACT	Meta-analysis, 15 studies (2,858 cases/2,907 controls): null result for NQO1; only

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								TNF-1031 was significant. Prior screening cited this PMID as "NQO1 genetic association meta-analysis" supporting the mechanism — that characterization is incorrect; the actual finding does not support an NQO1-PD genetic link.

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
11	[NEW — human PD-relevant target engagement] Oral oltipraz (200–400 mg/wk, 12 weeks)	failed to engage	Nrf2/phase-II pathway in human tissue (chronic smokers, lung epithelium)	OBSERVED_LITERATURE, CONTRADICTED	PubMed	PMID 15824161	ABSTRACT	"No increase in mRNA or enzymatic activity of phase II enzymes and no change in glutathione levels," despite 15% grade 2/3 (mostly GI) toxicity and all study withdrawals occurring in the oltipraz arms. Peripheral tissue, not

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								brain — but directly undercuts the assumption that oral dosing reliably drives oltipraz' s own proposed target engagement in humans at all. Surface d independently by the Critic, not the original reconstruction.
12	Oltipraz	has documented off-target activity via	LXR- α inhibition (antistatotic mechanism in	OBSERVED_LITERATURE, unaddressed	PubMed / Clinical Trials.gov	PMID 28225186 / NCT01373554	ABSTRACT	LXR- α is expressed in microglia and regulate

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
			NAFLD, same 30–120 mg/day dose range)	confounded				s CNS lipid/inflammatory tone — an unacknowledged confound for asserting the CNS effect is Nrf2-specific. Not evidence of harm, but a real gap in the mechanism's precision.

1.5 Agent-inferred bridges and their reasoning

Central bridge (Edge 6): Oltipraz's own confirmed CNS pharmacology (Edges 1–5: Nrf2 activation → NQO1/HO-1 + GPX4/SLC7A11/ACSL4 anti-ferroptotic program + anti-microglial effect, demonstrated in vivo in two independent CNS injury models) is bridged to PD-relevant dopaminergic neuroprotection (the same axis, shown protective for dopaminergic neurons in actual PD/MPTP models — but there driven by melatonin, schisanhenol, or β -caryophyllene, never oltipraz). Justification: the convergence of two independent oltipraz-specific studies (different investigators, different injury models) on the same Nrf2-driven signature suggests this is oltipraz's real, reproducible CNS pharmacology, and that same signature is independently

protective in dopaminergic-neuron contexts. Why it remains inference: no study has ever dosed oltipraz in any PD-relevant model; the CNS evidence is from acute nerve-injury and stroke paradigms in non-nigral tissue, not chronic degeneration in substantia nigra dopaminergic neurons — a real biological gap, not just an absence of data (Critic finding).

Secondary, weaker bridge (Edge 7): treated by the reconstruction agent as corroborating but shown by the Critic to be largely circular (re-indexed citations of the same primary paper plus PD-irrelevant cancer literature) — this strand should carry little independent evidentiary weight.

1.6 Unresolved relationships

- ClinicalTrials.gov independent re-verification of the "6 trials, all liver/lung, zero PD" finding: **SOURCE_ACCESS_FAILED** in the reconstruction pass (official [/api/v2/](#) blocked by robots.txt + org egress policy); the Critic independently reconfirmed the trial list via the same underlying data referenced in the N/P critic pass, but could not run a fresh independent CT.gov query this session either.
- Whether oltipraz achieves meaningful CNS/brain exposure at doses relevant to any of its human trials (all peripheral-indication trials; no CNS PK study located in the approved sources) — **UNRESOLVED**.
- Whether Nrf2 or NQO1 carry any curated Monarch gene-disease association with PD — **resolved as absent** (confirmed by direct query, not a search failure); this doesn't contradict the literature-based mechanism but means it cannot currently be corroborated through Monarch's curated layer.

1.7 Contradictory evidence

- PMID 25054016 does not support an NQO1-PD genetic association (null result) — corrects a screening mischaracterization (see Edge 10).
- The initial "FSP1–CoQ10 axis" framing has zero oltipraz-specific support — the real evidence is GPX4/SLC7A11/ACSL4-centric (Edge 3), a genuine mechanism revision.
- PMIDs 33271195 and 38455026, previously cited at screening as if supporting oltipraz, are actually about different compounds (β -caryophyllene, inflachromene) — target-class evidence only.
- **New (from Critic):** PMID 15824161 shows oral oltipraz failed to pharmacodynamically engage its own proposed Nrf2/phase-II target in a human trial, with a meaningful toxicity/withdrawal signal — this is real evidence against reliable human target engagement, independent of the CNS-penetration question.
- **New (from Critic):** the Monarch KG "corroborating strand" (NQO1-PRKN/AKT1 interactions) is largely circular, not independent evidence.
- No direct contradiction was found against oltipraz's core CNS pharmacology itself (Nrf2 activation, NQO1/HO-1/GPX4 induction, anti-microglial effect) — this compound-specific

evidence held up under independent re-verification by both the reconstruction agent and the Critic.

1.8 Independent Critic findings

All core PMIDs verified as real and accurately characterized; the reconstruction agent's self-corrections (dropping FSP1/CoQ10, nulling PMID 25054016, relabeling target-class-only citations) were accurate and appropriately conservative. The Critic's own independent contribution surfaced four new weaknesses not in the original reconstruction: (1) the oltipraz-specific CNS evidence comes only from acute-injury paradigms in non-nigral tissue — a real translational gap for a chronic, region-selective disease; (2) oral oltipraz failed to engage its own proposed target pharmacodynamically in the one human trial that measured it (PMID 15824161), with a real toxicity signal; (3) the Monarch KG "corroborating strand" is largely circular rather than independent; (4) NQO1's PD-stage-dependent expression (high early, absent late per PMID 31358653) raises a therapeutic-window concern. The Critic's overall read: the target-class hypothesis (Nrf2/antioxidant induction in PD) is legitimate, but the oltipraz-specific case is thinner than the reconstruction's framing conveyed.

1.9 Main remaining evidence gap

No study has ever tested oltipraz in any PD-relevant model (cellular, animal, or human), and the one human trial that measured whether oral oltipraz reliably engages its own proposed Nrf2 target found that it does not — at doses/routes not designed for or tested against CNS exposure. Whether oltipraz can achieve CNS target engagement at a tolerable dose, in a form relevant to chronic dopaminergic-neuron protection rather than acute injury response, is entirely open.

1.10 Final verdict

INCONCLUSIVE

1.11 Justification

Oltipraz's own CNS-relevant pharmacology (Nrf2 activation, NQO1/HO-1 induction, GPX4/SLC7A11/ACSL4 anti-ferroptotic engagement, anti-microglial suppression) is real, oltipraz-specific, and reproduced across two independent injury models — this is solid ground, not speculation. The link to PD specifically, however, rests entirely on an untested cross-model, cross-tissue extrapolation (acute nerve-injury/stroke pharmacology in non-nigral tissue → chronic dopaminergic-neuron protection), reinforced by a target-class literature (other compounds, same pathway, real PD models) rather than any oltipraz-specific PD data. This alone would support "plausible but unproven." What pushes it to INCONCLUSIVE rather than a stronger HOLDS_UP reading is a genuine negative signal the Critic surfaced: the one human

trial measuring oltipraz's own target engagement found it failed to occur, alongside a real toxicity/withdrawal signal — a concrete reason to doubt reliable target engagement independent of the (still entirely untested) blood-brain-barrier question. Nothing found actively falsifies the hypothesis (no evidence oltipraz harms dopaminergic neurons or opposes the proposed direction), so HYPOTHESIS_UNDERMINED is not warranted — but the evidence stops well short of confirming a viable path to PD-relevant efficacy.

2. PMX205

2.1 Candidate identity

PMX205 — cyclic hexapeptide, small-molecule antagonist of C5aR1 (CD88, complement C5a receptor 1), structurally related to PMX53. No brand name; still preclinical (zero registered clinical trials, independently reconfirmed twice this session via ClinicalTrials.gov's internal search API, since the public [/api/v2/](#) and [/search?](#) endpoints are robots-disallowed / egress-blocked on this network).

2.2 Proposed mechanistic hypothesis

α -Synuclein aggregation activates the classical complement pathway (C1q-initiated), proceeding through C3/C5 cleavage to cause complement-dependent cytotoxicity and, separately, microglial synaptic engulfment of dopaminergic circuitry in PD models. PMX205, a validated brain-active C5aR1 antagonist, is hypothesized to interrupt the C5a/C5aR1 arm of this signaling, dampening microglial/astrocyte activation — based entirely on PMX205's validated efficacy in four *other* CNS neuroinflammatory disease models (Alzheimer's, hypoxic-ischemic injury, intracerebral hemorrhage, radiation-induced cognitive decline), not on any PD or α -synuclein data point.

This mechanism is more precisely characterized, and more seriously undermined, than the original screening framing conveyed. Two independent investigations (reconstruction + Critic) converge on the same critical structural problem: the one PD-model paper that directly links complement to synaptic pruning implicates **C3-C3aR**, not C5-C5aR1 — PMX205 does not target the receptor associated with PD-model synaptic engulfment. The PD genetic model with the clearest synaptic-pruning phenotype (LRRK2 G2019S) implicates no complement component at all. And the one study that directly manipulates a complement gene in an α -synuclein-relevant model found that **removing C1q worsened**, not improved, the pathology.

2.3 Complete mechanism chain

1. α -Synuclein aggregation $\rightarrow$ activates classical complement pathway (C1q-initiated) — shown in α -syn-expressing cells, with C1q trend elevated in postmortem **MSA** (not PD) putamen
2. Classical pathway activation $\rightarrow$ C3/C5 cleavage $\rightarrow$ complement-dependent cytotoxicity (rescued in vitro by C3- and C5-specific inhibitors — but this data cannot distinguish the C5a/C5aR1 arm from the C5b-9 membrane-attack-complex arm)
3. C5 cleavage $\rightarrow$ generates C5a (canonical biochemistry, C5a is the C5aR1 ligand)
4. C5a–C5aR1 signaling $\rightarrow$ activates microglia/astrocytes (general CNS biology, GO-confirmed for C5AR1; zero curated PD-specific association)
5. Microglial/complement-linked synaptic engulfment of dopaminergic circuitry $\rightarrow$ shown in a **C3-C3aR-driven** rotenone PD toxin model (different receptor than PMX205's target) and, separately (no complement link at all), in a LRRK2 G2019S genetic PD model
6. PMX205 $\rightarrow$ antagonizes C5aR1 (well-established pharmacology, multiple independent labs)
7. PMX205/C5aR1 antagonism $\rightarrow$ suppresses microglial/astrocyte activation in four other CNS disease models (none is PD; none examines nigral/nigrostriatal circuitry)
8. **[Inference — the central, unresolved leap]** PMX205's C5aR1 blockade $\rightarrow$ hypothesized to dampen complement-driven dopaminergic-circuit engulfment in PD — **never tested; targets a different receptor than the one PD-model paper that shows complement-linked synaptic engulfment**

2.4 Edge-Level Evidence Ledger

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
1	α -Synuclein	activates	classical complement pathway (C1q)	OBSERVED_LITERATURE	PubMed	PMID 34399786	ABSTRACT	α -syn-expressing SH-SY5Y cells; C1q trend elevated specifically in MSA

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								postmortem putamen (not PD tissue). Paper frames PD/DLB/MSA together as synucleinopathies — treating the MSA data as strong PD-specific human evidence is a bigger inferential leap than the "synucleinopathy" framing suggests (MSA α -syn is predomi

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								nantly oligodendrocytic, pathologically distinct from PD's neuronal Lewy pathology).
2	Complement activation (C3/C5)	mediate s	cytotoxicity in α -syn-expressing cells	OBSERVED_LITERATURE	PubMed	PMID 34399786	ABSTRACT	Rescued by C5-inhibitor RaCl and C3-inhibitor Cp20 — C3/C5-dependent, not merely correlative. In vitro cell-line model only. RaCl blocks C5 cleavage

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								e itself, not the C5aR1 receptor — cannot isolate the C5a/C5aR1 arm from the C5b-9 (membrane-attack-complex) arm.
3	C5 cleavage	generates	C5a (C5aR1 ligand)	SUPPORTED_CONTEXT	Monarch	HGNC: 1338 (C5AR1), GO:0038178 / GO:0004878	MONARCH_KG	Canonical complement biochemistry (GO-annotated), general, not PD-specific.
4	C5aR1	activates	microglia / astrocytes	OBSERVED_KG	Monarch	HGNC: 1338, GO:0001774 / GO:004	MONARCH_KG	General biological-process annotated

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
						8143 / GO:0010759		on. Zero disease association attached — but this is a curation-coverage artifact, not a PD-specific signal: C5AR1 has zero curated gene-disease associations in Monarch for <i>any</i> disease, including Alzheimer's, despite ≥8 independent PubMed papers

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								linking PMX205/C5aR1 to AD models.
5a	Microglia	prune/engulf	dopaminergic fibers (LRRK2 G2019S PD genetic model)	OBSERVED_LITERATURE	PubMed	PMID 34409578	ABSTRACT (directly re-read by Critic)	Confirmed: abnormal microglial phagocytosis/pruning of DAergic fibers, inflammatory transcriptional remodeling, PD-related behavioral deficits. Zero mention of C1q/C3/C5/C5aR1 anywhere in this

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								paper — the clearest PD-model synaptic-pruning evidence available does not implicate complement at all.
5b	C3-C3aR axis	drives	synaptic engulfment / dark microglia (rotenone PD toxin model)	OBSERVED_LITERATURE (title only)	PubMed	PMID 41637879	ABSTRACT title (full text SOURCE_ACCESS_FAILED after repeated retries by both reconstruction and Critic)	Title confirms a complement-linked PD-model synaptic-engulfment finding — but implicates C3 and C3aR , a different

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								receptor system than C5/C5a R1. PMX205 does not target this receptor.
6	PMX205	antagonizes	C5aR1	OBSERVED_LITERATURE	PubMed	PMID 39147742, 41967875, 42570158, 32064396(†)	ABSTRACT (titles independently confirmed)	Well-established, reproducible pharmacology across independent labs. (†) PMID 32064396 (the PK study) could not be relocated by the Critic across 6

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								independent topical searches — flagged as a corroboration gap, not a fabrication.
7a	PMX205 (C5aR1 antagonism)	suppresses	inflammatory glial responses — Alzheimer's model	OBSERVED_LITERATURE	PubMed	PMID 39147742 (+ preprint duplicate 37662399)	ABSTRACT title	AD model, not PD. Critic independently surfaced ≥6 additional PMX205/C5aR1-AD-adjacent papers not in the original citation list — the pharmacological

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								I tool is on very solid ground generally.
7b	PMX205 (C5aR1 antagonist)	confers neuroprotection in	hypoxic-ischemic brain injury	OBSERVED_LITERATURE	PubMed	PMID 41967875	ABSTRACT title	Not PD; different brain regions/injury type than nigral degeneration.
7c	PMX205 (C5aR1 antagonist)	attenuates	neuroinflammation — intracerebral hemorrhage	OBSERVED_LITERATURE	PubMed	PMID 42570158	ABSTRACT title	Not PD; perihematomal region.
7d	C5aR1 inhibition	alleviates	cranial radiation-induced cognitive decline	OBSERVED_LITERATURE	PubMed	PMID 41085542 (peer-reviewed) = PMID 39005286 (bioRxiv preprint) — confirm	ABSTRACT title	Not PD; hippocampal/white-matter region. The "four independent CNS-disease

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
						ed same study, not two independent findings		models" is accurate as four <i>studies</i> , one (radiation) reported twice across preprint → publication.
8	PMX205	tested in	PD / α -synuclein model	CONTRADICTED (absence, independently reproduced twice)	PubMed	"PMX205 Parkinson," "PMX205 alpha-synuclein," "C5aR1 Parkinson" — all zero/near-zero on-topic hits	ABSTRACT (search listing)	Given how populated adjacent literatures are (PD-C1q, PD-C3aR, AD-C5aR1 each well-represented independently), this specific

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								intersection being empty reads as a genuine , searchable research gap, not mere obscurity (Critical finding).
9	C5AR1	associated with	Parkinson disease	CONTRADICTED (absence of KG support — but weak evidence, see Edge 4)	Monarch	HGNC: 1338 vs MONDO:0005180, total: 0	MONARCH_KG	PD's curated causal/correlated gene list in Monarch (SNCA, PRKN, PINK1, PARK7, LRRK2, DNAJC6, ATP13A2, FBXO7, SYNJ1,

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								PTPA, PLA2G6, CHCHD2, VPS13C, GBA1, MAPT, VPS35) contains no complement genes at all.
10	[NEW — direct contradicting evidence] Genetic C1q deletion	worsens	enteric α -synuclein neuropathology (while improving gut motility)	CONTRADICTED — directly complicates the core premise	PubMed	PMID 37961460 (preprint, bioRxiv)	ABSTRACT	The one study that directly manipulates a complement gene in an α -synuclein-relevant model found complement engulfment/exhaustion

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								of ENS-resident macrophages was protective/clearance-oriented for neurons, not purely pathogenic — a real dissociation. Peripheral (enteric), not central, so generalizability to nigral circuits is uncertain, but it directly challenges the "complement

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								activation = bad, block it = good" premise underlying the PMX205 hypothesis.

2.5 Agent-inferred bridges and their reasoning

Bridge A (Edge 3): general complement biochemistry (C5→C5a→C5aR1) applied to the α-synuclein/complement-cytotoxicity finding (Edge 2) — reasonable canonical biochemistry, but PMID 34399786's C5-inhibition data cannot itself distinguish the C5a/C5aR1 signaling arm from the C5b-9 membrane-attack-complex arm, so this is inference from general pathway structure, not a demonstrated finding.

Bridge B (Edge 8, the central and unavoidable leap): bridges (i) real synucleinopathy-model evidence that α-syn triggers complement-dependent cytotoxicity (Edges 1–2, MSA tissue + cell-line data), (ii) KG-confirmed general biology that C5aR1 activates microglia (Edge 4), (iii) PD-model evidence of complement-linked synaptic engulfment — but via a *different receptor*, C3aR, not C5aR1 (Edge 5b), and separately, complement-silent synaptic pruning in a genetic PD model (Edge 5a), and (iv) PMX205's validated efficacy suppressing C5aR1-driven neuroinflammation in four non-PD CNS diseases (Edges 6–7d). No single study connects these; no PD model, no α-synuclein model, and no C5aR1-specific tool (genetic or PMX205) have ever been combined. Justified as a plausible, testable hypothesis by the convergence of independently real findings — but the receptor-level mismatch between PMX205's actual target and the one complement receptor shown relevant to PD-model synaptic pruning is a specific, previously underweighted structural weakness in this bridge, not just a general absence of data.

2.6 Unresolved relationships

- PMID 32064396 (PK study, cited as evidence of CNS-penetrant peripheral dosing) — could not be relocated by the Critic in 6 independent topical searches; **UNRESOLVED**, not fabricated, but not independently corroborated either.

- PMID 41637879 full text (the C3-C3aR rotenone PD-model paper) — **SOURCE_ACCESS_FAILED** after repeated retries by both agents; only the title is confirmed.
- ClinicalTrials.gov: public [/api/v2/](#) and [/search?](#) endpoints blocked (robots.txt + org egress policy); the permitted [/api/int/](#) endpoint was used instead and returned an authoritative **total: 0** for PMX205, independently reconfirmed — treated as resolved via an alternate access *method* for the same approved source, not a substitution.
- Google Patents' independent confirmation of PMX205's patent-novelty status remains outside this Deep Validation's approved-source scope (PubMed/Monarch/CT.gov only) and stays an open item from the earlier N/P critic.

2.7 Contradictory evidence

- The clearest PD-model synaptic-pruning paper (LRRK2 G2019S, PMID 34409578) does not implicate complement at all — weakens, without falsifying, the premise that PD-relevant synaptic pruning is complement-driven.
- The one PD-model paper that does link complement to synaptic engulfment implicates **C3-C3aR**, a different receptor than PMX205's target, C5aR1 (PMID 41637879, title-level).
- **PMID 37961460: genetic C1q deletion worsened enteric α -synuclein pathology** — direct evidence complicating the simple "complement=bad" framing in the one study that directly manipulates a complement gene in an α -synuclein-relevant model.
- Monarch shows zero curated PD-C5aR1 association — but this is shown to be a weak signal on inspection, since the same zero-association result holds for Alzheimer's disease despite abundant AD-C5aR1 literature (curation-coverage artifact, not a PD-specific finding).
- All originally-cited PMIDs were confirmed real and accurately titled; no fabricated or misrepresented citations were found by either agent.

2.8 Independent Critic findings

The Critic independently verified nearly every cited PMID and found PMX205's core pharmacology (C5aR1 antagonism, multi-model CNS anti-inflammatory efficacy) even better-supported than originally cited, surfacing ≥ 8 additional corroborating papers. But the Critic's own contribution was to sharpen the central weakness: the receptor-level mismatch between PMX205 (C5aR1) and the one PD-model complement-pruning paper (C3aR) is a structural problem, not just "PMX205 untested in PD" — the one complement receptor with any PD-model synaptic-pruning association is not the one PMX205 targets. The Critic also found that none of PMX205's four supporting CNS-disease efficacy models examine the substantia nigra / nigrostriatal circuitry central to PD, and that the Monarch "zero PD association" is a weaker signal than presented (same zero result for AD, a curation-coverage issue). The Critic's overall read: a defensible, testable hypothesis, but resting on a longer and shakier inferential chain — spanning a cross-disease (MSA $\rightarrow$ PD) tissue extrapolation, a receptor-mismatch in the

one PD-relevant complement finding, and cross-region (non-nigral) efficacy data — than the reconstruction's confident framing conveyed.

2.9 Main remaining evidence gap

No study has combined PMX205, or any C5aR1-specific genetic or pharmacological tool, with a PD or α -synuclein model. The one complement receptor independently linked to PD-model synaptic engulfment (C3aR) is not the receptor PMX205 blocks (C5aR1) — whether C5aR1 blockade would meaningfully affect PD-relevant dopaminergic-circuit pathology, given this receptor-specific gap, is entirely untested.

2.10 Final verdict

HYPOTHESIS_UNDERMINED

2.11 Justification

PMX205 itself is a well-validated, CNS-active C5aR1 antagonist with genuinely broad and reproducible anti-neuroinflammatory efficacy — that part of the case is solid. But the specific mechanistic bridge to PD is undermined by more than an absence of data: the one PD-model study that directly links complement activation to synaptic engulfment implicates a *different* complement receptor (C3aR) than the one PMX205 targets (C5aR1); the PD-model paper with the clearest synaptic-pruning phenotype (LRRK2 G2019S) implicates no complement component at all; and the one study that directly manipulates a complement gene in an α -synuclein-relevant context found that removing complement (C1q) *worsened* rather than improved the pathology — directly complicating the "block complement, protect neurons" premise the hypothesis depends on. None of PMX205's four supporting efficacy models examines the nigral/nigrostriatal circuitry central to PD, and the human-tissue evidence for α -synuclein-complement activation (MSA putamen) comes from a related but pathologically distinct synucleinopathy, not PD itself. Taken together — receptor-level mismatch, complement-silent PD-model pruning evidence, and a directly contradicting genetic finding — this is not simply an untested-but-plausible extrapolation; specific, real evidence weighs against the proposed mechanistic bridge as currently framed, even though PMX205's own pharmacology and the general complement-neuroinflammation-PD research area remain legitimate.

Explicit note on conflicts with prior screening (per instruction: do not silently overwrite)

- **Oltipraz:** initial screening's M3/F2/Direction=consistent/CNS=feasible ratings, and its citation of PMID 25054016 as NQO1 genetic-association support, are **not fully supported** by this Deep Validation pass. The FSP1-CoQ10 mechanism framing is revised (no oltipraz-specific support found); PMID 25054016 is a null result, not positive evidence; and new evidence (PMID 15824161) shows oral oltipraz failed human target engagement in the one trial that measured it. These are presented as revisions arising from independent verification, not as a rejection of the underlying retention decision — the critic-vetted N3/N4, P4 novelty status is unaffected and still stands.
- **PMX205:** initial screening's Direction=consistent and CNS=feasible ratings are **challenged** by this Deep Validation pass. New evidence not present in the initial screening — the C3aR-not-C5aR1 receptor mismatch in the one PD-model complement-pruning paper, and the C1q-deletion-worsens-pathology finding — was not available to or considered in the original screening and materially weakens the mechanistic case, independent of the (unaffected) N3/P4 novelty status.

3. NEFLAMAPIMOD (VX-745)

3.1 Identity

Selective, ATP-competitive p38 α /MAPK14 kinase inhibitor (PMID 36130946). No Monarch Initiative entity exists for "neflamapimod" or "VX-745" (confirmed by both mechanism and critic agents). CAS/CID UNRESOLVED (not indexed in Monarch).

3.2 Hypothesis (as carried into this validation)

p38 α inhibition, by relieving a Rab5/endosomal-trafficking or kinesin/axonal-transport bottleneck, could rescue a degenerating neuronal population in PD, on analogy with neflamapimod's demonstrated effect on basal forebrain cholinergic neurons (BFCNs) in dementia with Lewy bodies (DLB).

3.3 Mechanism chain — REVISED from initial screening

initial screening framed this as "axonal transport/microtubule-motor dysfunction," citing p38 MAPK's general link to PD dopaminergic pathology. This validation's evidence requires two revisions:

1. **The only mechanistically detailed, compound-specific evidence is p38 α →Rab5→BFCN rescue, and it is entirely in DLB, not PD** (PMID 36130946, *Nat Commun* 2022 — mechanistic paper; NCT04001517, human Phase 2a DLB trial — corrected from screening-stage "Phase 2B" mischaracterization, independently reconfirmed by the critic via PMID 38706271's own title, which distinguishes "Phase 2A" NCT04001517 from "Phase 2B" RewinD-LB/NCT05869669; PMID 37657939, *Neurology* 2023 — plasma p-tau181 used as a patient-*stratification* biomarker, not a pharmacodynamic target-engagement readout, confirmed verbatim by the critic).
2. **The separate, genuinely PD-specific paper (PMID 40328804, *NPJ Parkinson's Disease* 2025) supports an AMPK/p38-MAPK→kinesin-impairment axis in dopaminergic neurons exposed to α -synuclein fibrils, but does not test neflamapimod, and the specific p38 isoform (α vs. $\beta/\gamma/\delta$) could not be confirmed by either agent** — recorded UNRESOLVED, not assumed to be MAPK14/p38 α .

The critic's independent search surfaced two on-topic papers the reconstruction missed — **PMID 37163617** (*Sci Transl Med* 2023, "Inhibition of p38 α MAPK restores neuronal p38 γ MAPK and ameliorates synaptic degeneration in a mouse model of DLB/PD") and **PMID 26517532** (*Cell Death Differ* 2016, "c-Abl-p38 α signaling plays an important role in MPTP-induced neuronal death") — both explicitly isoform-specific (p38 α) and both PD-adjacent, but full text could not be retrieved by the critic to confirm whether neflamapimod itself, dopaminergic pathology, or motor outcomes were tested. **Net effect: the isoform-specificity gap is**

narrower than the reconstruction reported, but not closed — this is an explicit, tracked revision made mid-critique, not a silent correction.

3.4 Edge-level evidence ledger (key edges; full ledgers preserved in both source agent transcripts)

#	Subject → Object	Status	Source	Statement
1	Neflamapimod → p38α/MAPK14 (ATP-competitive inhibitor)	OBSERVED_LITERATURE	PMID 36130946	Confirms identity and target
2	p38α inhibition → Rab5 normalization → BFCN rescue	OBSERVED_LITERATURE	PMID 36130946 (<i>Nat Commun</i> 2022)	Core mechanistic paper — DLB model, not PD
3	Neflamapimod → clinical improvement	OBSERVED_LITERATURE	NCT04001517; PMID 38706271	Phase 2a DLB trial (corrected trial-phase label)
4	Plasma p-tau181 <2.2 pg/mL → greater cognitive/motor/functional response	OBSERVED_LITERATURE (verified verbatim by critic)	PMID 37657939	Patient-stratification biomarker, NOT pharmacodynamic target engagement
5	AMPK/p38-MAPK → kinesin impairment → PD dopaminergic pathology (α-syn fibril exposure)	OBSERVED_LITERATURE, isoform UNRESOLVED	PMID 40328804	Real PD-specific paper; does not test neflamapimod; isoform unconfirmed by either agent
6	MAPK14 (gene) → Parkinson disease	UNRESOLVED (zero KG association)	Monarch	Zero curated MAPK14-PD gene-disease edge. Critic

#	Subject → Object	Status	Source	Statement
				independently found this is ALSO zero for Lewy body dementia (Mn-neflamapim od's actual efficacy indication) — tempering how much the PD-specific zero should be read as a red flag, since Monarch's curation lags clinical evidence generally
7	Mouse Mapk14 knockout → phenotypes	Mostly UNRESOLVED, one CONTRADICTED sub-claim	Monarch, PMID 26377476	Reconstruction stated "none of 84 phenotypes are nervous-system/ motor-related"; critic independently found and confirmed via raw KG citation a genuine cognitive/behavioral phenotype ("abnormal associative learning," PMID 26377476) attributed to Mapk14 —

#	Subject → Object	Status	Source	Statement
				directly contradicts the literal "none...nervous-system-related" claim, though it is cognitive, not motor (the "motor" half of the original claim held up under critic re-scan of ~97 associations)
8	Germline Mapk14 knockout mice	Confound, not itself an edge	Monarch (critic finding)	Full germline knockouts are embryonic/perinatal lethal (placental/cardiac failure) — the "no CNS/motor phenotype" finding is a ceiling effect of lethality, not demonstrated absence of CNS relevance. Not flagged by the reconstruction agent.
9	Neflamapimod → registered clinical trials	OBSERVED_LITERATURE	CT.gov (/api/int/studies)	8 total trials — 3 DLB, 2 Alzheimer's, 1 PPA, 1 stroke motor-recovery, 1 Huntington's (terminated).

#	Subject → Object	Status	Source	Statement
				Zero PD trials. Independently re-confirmed by critic.
10	"neflamapimod AND Parkinson" / "neflamapimod AND dopaminergic"	UNRESOLVED (zero-result, reconfirms N3)	PubMed	16 hits / 0 hits respectively (critic's independent re-run) — entirely DLB/Lewy-body-spectrum literature, no idiopathic-PD-specific compound study
11	PMID 26261796, PMID 38412454	SOURCE_ACCESS_FAILED (both agents, 5+ and 8+ retries respectively)	—	Genuinely unresolved, not assumed to support or contradict. Critic partially recovered PMID 38412454's title — "Targeting Lewy body dementia with neflamapimod-rasagiline hybrids" (medicinal-chemistry hybrid-molecule paper, DLB-focused) — but could not access the abstract body to determine if any

#	Subject → Object	Status	Source	Statement
				PD/dopaminergic model was used.
12	PMID 37163617, PMID 26517532 (isoform-specific p38α, PD-adjacent)	UNRESOLVED — found but unverifiable	PubMed (title-only)	New papers surfaced by critic search; not in original reconstruction; full text inaccessible to confirm relevance to neflamapimod itself

3.5 Inferred bridges + reasoning (AGENT_INFERENCE)

- **DLB → PD extrapolation:** neflamapimod's only clinical/mechanistic dataset is in DLB (BFCN, cholinergic). Extrapolating to PD dopaminergic neuroprotection requires assuming p38α-driven pathology generalizes across neuron types and diseases — an inference, not evidenced. Both agents independently reproduced "zero direct PD/dopaminergic evidence for this compound" via distinct searches.
- **BFCN → dopaminergic-neuron bridge:** no mechanistic study connects Rab5-dependent BFCN rescue to dopaminergic-neuron biology. This gap is unaddressed by any evidence in either pass — a genuine, unclosed extrapolation, not merely an unresolved search.
- **p38-generic → p38α-specific PD relevance:** narrower than the reconstruction reported (two additional title-level hits found), but not closed — full text needed to confirm isoform and whether neflamapimod itself was tested in either paper.

3.6 Unresolved relationships

- Whether PMID 37163617 and PMID 26517532 test neflamapimod itself, dopaminergic/nigrostriatal pathology, or motor outcomes (full text inaccessible to both agents).
- The functional significance of the germline-lethality confound on Monarch's "no CNS/motor phenotype" claim — no conditional/adult-viable Mapk14 knockout data were located.

- Whether DCTN4(dynactin)/KIF22(kinesin) PPI edges in Monarch are causal or purely correlative associations (flagged, not resolved, by the reconstruction).

3.7 Contradictory evidence

One genuine, independently-verified partial contradiction: Monarch KG lists a real nervous-system/behavioral phenotype ("abnormal associative learning," PMID 26377476) for mouse Mapk14 — directly contradicting the reconstruction's literal claim that none of the 84 recorded phenotypes are nervous-system-related. Scope-limited (cognitive, not motor; a single occurrence among ~97 associations; primary text of PMID 26377476 itself inaccessible to confirm the experimental context) — **CONTRADICTED** for the narrow literal claim, not for the broader "no direct PD evidence" conclusion, which both agents independently reproduce.

3.8 Critic findings (summary — full report in agent transcript, agentId [aa8f75ed8244c150c](#))

The critic reproduced the reconstruction's central "zero direct PD/dopaminergic evidence for neflamapimod" finding via independent search, and added: (a) the Mapk14-phenotype contradiction above; (b) the germline-lethality confound on the "no CNS phenotype" claim; (c) two under-verified but potentially isoform-narrowing PMIDs the original search missed; (d) confirmation that the screening-stage "Phase 2B" mischaracterization was correctly caught and fixed by the reconstruction. Critic's own words: *"None of these findings rescue a PD-specific mechanistic case for neflamapimod — if anything, the corrected picture is messier and less clean than the reconstruction presented... while the core verdict — no direct PD/dopaminergic evidence for this specific drug exists in any approved source — holds up under independent adversarial testing."*

3.9 Main evidence gap

No study has ever tested neflamapimod in a PD, dopaminergic-neuron, or nigrostriatal model of any kind. The entire evidentiary case rests on a same-target (p38 α), different-disease (DLB), different-neuron-population (BFCN, cholinergic) dataset, bridged to PD entirely by inference.

3.10 Explicit conflicts/revisions relative to initial screening / N/P critic

- the screening-stage trial-phase label ("Phase 2B") is corrected to Phase 2a (NCT04001517); RewinD-LB (NCT05869669) is the actual Phase 2B trial — initial screening conflated the two.
- initial screening characterized M3/F3 mechanistic/functional support without stating plainly that essentially all of it is DLB-specific, not PD-specific — this Deep Validation makes that qualification explicit and load-bearing rather than implicit.
- the screening-stage framing of p38 MAPK-PD linkage as reasonably well-established is revised downward: the one genuinely PD-specific paper (PMID 40328804) does not test neflamapimod and cannot even confirm the isoform.

3.11 Final verdict: **INCONCLUSIVE**

3.12 Justification

Neflamapimod has zero direct PD or dopaminergic-neuron evidence, in either direction — no study has tested it in a PD model, and no contradicting evidence was found that would flatly undermine the target's biological plausibility. What exists is a coherent, real mechanism (p38 α →Rab5→BFCN rescue) validated in a different but neurologically adjacent disease (DLB), with a genuine but unclosed translation gap to PD dopaminergic neurons. This is not a hypothesis that "holds up" (it has never been tested in the disease of interest) nor one that is "undermined" (no material contradicting evidence was found for the PD-specific claim itself — the one contradiction found is a narrow, non-fatal KG-labeling issue). It remains a biologically plausible but entirely untested repurposing hypothesis, appropriately classified **INCONCLUSIVE** pending a direct PD/dopaminergic-model test.

4. EMRICASAN (IDN-6556)

4.1 Identity

Pan-caspase inhibitor targeting caspase-1/3/8/9 (irreversible). 94 PubMed records for "emricasan," 64 for "IDN-6556," confirmed same compound. Zero Monarch KG entity exists for emricasan (independently reconfirmed by critic).

4.2 Hypothesis (as carried into this validation)

Pan-caspase inhibition by emricasan could protect dopaminergic neurons from apoptotic death in PD, either via a Golgi-fragmentation-rescue mechanism (the original screening framing) or via broader anti-apoptotic neuroprotection (this validation's revised framing).

4.3 Mechanism chain — REVISED from initial screening

screening-stage caspase-2/3→GRASP65/giantin→Golgi-fragmentation framed ~~discarded~~: the real PD-model Golgi-fragmentation literature (PMID 33078843, PMID 37855309) attributes the phenotype to Rab1/syntaxin-5 dysregulation, with zero mention of caspases, GRASP65, or

giantin in either paper; a direct connecting search ("Rab1 AND caspase AND Golgi") returned nothing relevant (UNRESOLVED per the zero-result rule, not CONTRADICTED). A screening citation misattribution is also corrected: the real caspase-mediated-giantin-cleavage evidence is **PMID 14970262**, not whatever PMIDs were originally cited during screening.

Revised primary hypothesis: emricasan's directly-evidenced pharmacology is broad pan-caspase inhibition protecting neural cells in **non-PD acute-injury** contexts — Zika-infected cortical progenitors (PMID 27571349), rat cerebral ischemia-reperfusion (PMID 29103102), TMAO/PANoptosis hippocampal model (PMID 41249794, **combination-therapy only**, not emricasan monotherapy) — extrapolated via **AGENT_INFERENCE** to PD dopaminergic apoptosis. The novelty claim is also refined: "emricasan AND Parkinson disease" is not a genuine zero — it returns 1 hit, **PMID 31612876**, an assay-validation paper using emricasan as a caspase-3-inhibitor control compound in an MPP⁺ PD cellular model, not a therapeutic study.

4.4 Edge-level evidence ledger (key edges)

#	Subject → Object	Status	Source	Statement
1	Emricasan → caspase-1/3/8/9	OBSERVED_LITERATURE	94/64 PubMed records	Identity confirmed
2	"emricasan AND Parkinson disease"	UNRESOLVED — not a true zero	PMID 31612876	1 hit; assay-validation/control-compound use in MPP ⁺ model, not therapeutic
3	PD-model Golgi fragmentation → Rab1/syntaxin-5 (not caspase/GRASP 65/giantin)	OBSERVED_LITERATURE	PMID 33078843, PMID 37855309	Zero caspase/GRASP 65/giantin mention in either paper — screening mechanism chain does not connect to real PD literature
4	Caspase-mediated giantin cleavage	OBSERVED_LITERATURE (corrected citation)	PMID 14970262	Real source for this claim; cited in screening

#	Subject → Object	Status	Source	Statement
				the wrong PMIDs
5	Emricasan → protection in non-PD acute neural injury	OBSERVED_LITERATURE	PMID 27571349 (Zika), PMID 29103102 (ischemia-reperfusion), PMID 41249794 (TMAO/PANoptosis, combo-only)	Real but non-PD, and one of three is combination-therapy-only — should carry near-zero standalone weight (both agents agree)
6	DA-neuron-specific genetic Casp3 knockout → MPTP model outcome	CONTRADICTED (relative to apoptosis-inhibition-as-protective logic)	PMID 39223107 (<i>Cell Death Dis</i> 2024)	No net neuroprotection; death mode switches apoptosis→necrosis; worse microglial response; authors explicitly conclude apoptosis-inhibition is not a beneficial PD strategy
7	CASP1/CASP3/CASP8/CASP9 → PD (gene-disease)	UNRESOLVED (zero KG association)	Monarch	Independently reconfirmed by critic: CASP3/1/9 have zero disease association of any kind in Monarch (not just zero PD); CASP8 has 2

#	Subject → Object	Status	Source	Statement
				non-PD associations
8	GORASP2 (GRASP65), GOLGB1 (giantin), RAB1A → PD or any disease	UNRESOLVED (zero KG association)	Monarch	Independently reconfirmed by critic — the discarded screening mechanism chain has zero KG support at every node, same as the revised chain
9	Caspase inhibitor (any) → PD clinical trials	UNRESOLVED (zero)	CT.gov /api/int/studies	"query.cond=Parkinson+Disease & query.term=caspase" = 0 studies — no caspase-targeted compound of any kind has reached PD clinical development, independently confirmed by critic
10	Emricasan → PD clinical trials	UNRESOLVED (zero)	CT.gov	18 total emricasan trials, all NASH/liver/COVID/islet-transplant/PK — none PD
11	SNCA (positive control) → PD	OBSERVED_KG	Monarch	3 causal associations — confirms the KG

#	Subject → Object	Status	Source	Statement
				query method is valid and the caspase-family zero-results are real findings, not broken queries

4.5 Inferred bridges + reasoning (AGENT_INFERENCE)

- **Acute injury → chronic neurodegeneration**: emricasan's real neuroprotective data is in acute insult models (Zika, ischemia-reperfusion); extrapolating to PD's chronic, progressive dopaminergic loss is a distinct, untested inferential leap.
- **Non-PD tissue → nigral dopaminergic vulnerability**: none of the three supporting papers use PD-relevant tissue or dopaminergic neurons specifically.
- **Pharmacological pan-inhibition → genetic single-caspase perturbation**: the one PD-relevant functional test (PMID 39223107) used genetic, single-caspase (Casp3-only) knockout, not pharmacological pan-caspase inhibition — a real but, per the critic's stress-test, weakly-grounded basis for hedging away the contradiction (see 2.7/2.8).

4.6 Unresolved relationships

- Whether PMID 31612876's use of emricasan as an assay control incidentally reports any efficacy-relevant signal in the MPP⁺ system (flagged as an open risk by the critic — reconstruction's "just a control" framing may understate this; neither agent could access full text to resolve it).
- Whether caspase-8's role in gating necroptotic/pyroptotic death (relevant to the critic's stress-test of the pan-caspase hedge) is confirmed in the primary literature — the critic's reasoning here is explicitly flagged as unverified logical inference, not sourced fact.

4.7 Contradictory evidence

PMID 39223107 stands as the central contradicting data point: DA-neuron-specific genetic Casp3 knockout in the MPTP PD mouse model produced no net neuroprotection, switched cell death from apoptosis to necrosis, and worsened microglial/neuroinflammatory response; authors explicitly conclude apoptosis-inhibition is not a beneficial PD strategy. The reconstruction's hedge (genetic-vs-pharmacological; single-caspase-vs-pan-caspase) was stress-tested by the critic and found **not mechanistically grounded** — and the pan-caspase point specifically may cut the *wrong* way: if caspase-8 normally gates necroptotic/pyroptotic death, a pan-caspase inhibitor removing caspase-8 activity alongside caspase-3 could plausibly

expose the *same or worse* necrotic/lytic failure mode, not a smaller one. This reasoning is explicitly flagged by the critic as **AGENT_INFERENCE** on an unverified point, not as new sourced fact — but it means the contradiction should weigh **more heavily** than the reconstruction's original hedge implied.

4.8 Critic findings (summary — full report in agent transcript, agentId **a19497db8943e841f**)

The critic could not independently access any cited PMID's full text this session (PubMed CAPTCHA-walled; E-utilities and the NCBI Literature Citation Exporter both robots-disallowed; direct API access blocked at network egress) — an important, explicitly disclosed limitation. Working from CT.gov and Monarch (both fully accessible) plus logical stress-testing of the reconstruction's own stated claims, the critic found: (1) the entire mechanism chain — both discarded and revised versions — has **zero KG-level support at every single node**, a broader and more damaging gap than "no CASP3-PD edge" alone; (2) zero PD clinical trial history for emricasan or for any caspase inhibitor as a class; (3) the PMID 39223107 hedge does not survive scrutiny and may run backwards; (4) the combination-therapy-only status of PMID 41249794 means the "three supporting studies" tally is inflated — effectively two independent monotherapy data points, not three. Critic's own words: *"Given the instruction not to rescue the hypothesis reflexively, and given that everything I could independently confirm points toward 'weaker than presented' rather than 'stronger,' I record the verdict as UNDERMINED rather than merely INCONCLUSIVE."*

4.9 Main evidence gap

No pharmacological caspase inhibitor — emricasan or otherwise — has ever been tested in a PD model, and the one genetic proxy experiment that exists (Casp3 knockout, MPTP model) argues against the core logic (apoptosis-inhibition as neuroprotective) rather than for it.

4.10 Explicit conflicts/revisions relative to initial screening / N/P critic

- screening-stage caspase→GRASP65/giantin→Golgi-fragmentation mechanism chain is discarded as unsupported by the actual PD Golgi-fragmentation literature (which is Rab1/syntaxin-5-driven).
- screening-stage citation for caspase-giantin cleavage is corrected to PMID 14970262.
- screening-stage "emricasan AND Parkinson disease = 0 results" novelty claim is corrected — it is 1 result (PMID 31612876), though this does not overturn the N3/P4 conclusion since the hit is a non-therapeutic assay-control use.
- initial screening characterized F2 evidence via PMID 29103102 (ischemia-reperfusion) without noting PMID 41249794 is combination-only — this Deep Validation explicitly discounts that data point's independent weight.

— N3/P4 was retained after N/P critic reassessment and was independently re-examined during this validation.

4.11 Final verdict: **HYPOTHESIS_UNDERMINED**

4.12 Justification

The one direct functional test most analogous to emricasan's proposed PD mechanism — genetic ablation of a caspase emricasan inhibits, in dopaminergic neurons, in a PD mouse model — found no benefit and a worse inflammatory outcome, directly contradicting the apoptosis-inhibition-as-neuroprotective logic underlying the whole hypothesis. Every node in both the original and revised mechanism chains is independently confirmed to have zero curated disease-association support in Monarch, no caspase-targeted compound of any class has reached PD clinical trials, and the compound's supporting evidence is confined to acute, non-PD, largely non-dopaminergic injury contexts (one of which is combination-therapy-only and should not be counted independently). The critic's stress-test found the reconstruction's own hedge against the central contradiction does not hold up mechanistically and may understate the risk. This is a materially undermined hypothesis, not merely an unproven one.

5. ANX005 (Tanruprubart)

5.1 Identity

Humanized anti-C1q neutralizing monoclonal IgG4 antibody (Annexon), blocking the classical complement pathway upstream of C3 cleavage. Confirmed via PMID 29202623 ("Nonclinical Development of ANX005...") and NCT07020819's official trial title linking ANX005/Tanruprubart.

5.2 Hypothesis (as carried into this validation)

Systemic C1q neutralization by ANX005 blocks classical-complement-initiated, microglia-mediated synaptic pruning of dopaminergic synapses in PD, slowing synaptic/neuronal loss — requiring (a) C1q itself (not just downstream C3/C3aR) being a functionally significant PD driver, (b) net benefit from C1q loss-of-function, and (c) meaningful CNS/CSF target engagement by a full-length IgG4.

5.3 Mechanism chain — REVISED from initial screening, then further complicated by the critic

initial screening found no PD- or C1q-specific PD literature directly tying ANX005's target to PD pathology. This validation's reconstruction found genuine upgrades — PD-specific human postmortem C1q evidence (PMID 42481480, 42477211), a Parkinsonism-model paper with direct functional C1q-neutralization rescue of dopaminergic synapses (PMID 42531937), and direct CSF target-engagement data for ANX005 itself in a human trial (PMID 41787606) — alongside a genuine complication: **genetic C1q deletion worsens histopathology (while improving function) in a PD-relevant enteric α -synuclein model** (PMID 41554750, peer-reviewed *Nat Communications* 2026, confirming preprint PMID 37961460) — a direct hit on ANX005's own target, independently re-verified by the critic via a live PubMed search that reproduced the exact title, author list, and journal.

The critic's independent work sharpened this picture further and, on balance, tilted it more unfavorable:

1. The **sole positive functional-rescue citation (PMID 42531937)** — the only paper offering direct evidence that neutralizing C1q protects dopaminergic synapses in a Parkinsonian model — could not be independently verified by the critic either (full text inaccessible to both agents across the whole pipeline). This is now flagged as the single most load-bearing, least-verified citation in the entire hypothesis.
2. The critic independently pulled the raw ClinicalTrials.gov registry record for ANX005's own HD trial (NCT04514367) — separately from the journal paper — and found **CSF complement-activation product C4a rose (not fell) during ANX005 dosing** (+38.18 $\mu\text{g/L}$ at Week 24, +23.28 $\mu\text{g/L}$ at Week 36), and **CSF neurofilament light chain (NfL) rose** (+318 to +509 ng/L) — while blood NfL trended slightly down. This is new evidence, not in the reconstruction, and is directionally unfavorable: if C1q neutralization were functionally suppressing classical-pathway activation in the CNS, C4a should fall; a rising C4a alongside rising CSF NfL suggests target binding/saturation did not translate into functional pathway blockade or reduced neurodegeneration markers in this trial.
3. The critic also found the PMID 41554750 title itself specifies the contradicting finding is in **male mice only** — a real scope limitation neither agent had previously flagged, though PD's own male-skewed incidence means this does not obviously rescue the hypothesis either.

5.4 Edge-level evidence ledger (key edges)

#	Subject → Object	Status	Source	Statement
1	ANX005 → C1q (binds/neutralizes)	OBSERVED_LITERATURE	PMID 29202623	Identity/target confirmed
2	ANX005 → CSF/serum target saturation (HD trial)	OBSERVED_LITERATURE	PMID 41787606	"Full saturation... in serum and the CSF" — compound-level, HD not PD
3	ANX005 (HD trial) → CSF C4a, CSF NfL trends	New, critic-sourced, unfavorable	CT.gov NCT04514367 (raw registry, independent of the paper)	C4a and NfL both rose during dosing — cuts against "functional pathway blockade" and "clinical activity" framing
4	Aggregated α-synuclein → binds/activates C1q	OBSERVED_LITERATURE (reconstruction only; SOURCE_ACCESS_FAILED for critic)	PMID 42477211	Human PD SN tissue + rat PFF model; not independently re-verified by critic
5	Microglial cGAS-STING-C1q axis → dopaminergic synapse loss, motor dysfunction (Mn-Parkinsonism)	OBSERVED_LITERATURE (reconstruction only; SOURCE_ACCESS_FAILED for critic — the single most load-bearing citation,	PMID 42531937	Sole functional-rescue paper for C1q specifically; toxin model, not synucleinopathy; genetic-vs-pharmacological neutralization

#	Subject → Object	Status	Source	Statement
		unverified by either agent's full-text access)		method unconfirmed
6	Genetic C1q deletion → worsens enteric histopathology, improves gut function (PD-relevant PFF model, male mice)	CONTRADICTED (direct hit on ANX005's own target)	PMID 41554750 (peer-reviewed; confirms preprint 37961460)	Independently re-verified verbatim by critic via live search — title, authors, journal, key sentence all confirmed
7	C3/C3aR (not C1q) → causally rescues MPTP-model deficits	OBSERVED_LITERATURE (reconstruction; SOURCE_ACCESS_FAILED for critic this validation)	PMID 42263400	C1q measured/elevated but not functionally tested; rescue experiments target C3/C3aR only — reproduces the PMX205-branch receptor-mismatch pattern
8	C5/C3 inhibitors (not C1q) → rescue synucleinopathy toxicity	OBSERVED_LITERATURE	PMID 34399786	In vitro + MSA (not PD) tissue
9	C1QA (human gene) → PD (gene-disease)	UNRESOLVED (zero KG association)	Monarch	Independently reconfirmed by critic: total:

#	Subject → Object	Status	Source	Statement
				0; PD's own 14 causal + 32 correlated genes in Monarch contain no complement genes at all — second independent confirmation of this pattern (first was C5AR1 in the PMX205 validation)
10	ANX005 → PD clinical trials	UNRESOLVED (zero)	CT.gov	7 total trials (ALS, wAIHA, healthy volunteers, 2×GBS, HD, FORWARD/GBS) — zero PD, independently reconfirmed by critic, no new trials since screening
11	C1QC expression → precedes inhibitory synapse loss in early PD-spectrum (iLBD) tissue	OBSERVED_LITERATURE (reconstruction only; SOURCE_ACCESS_FAILED for critic)	PMID 42481480	Human postmortem, correlative not causal; not independently re-verified

5.5 Inferred bridges + reasoning (AGENT_INFERENCE)

- **Mn-Parkinsonism** → **synucleinopathy PD convergence**: the reconstruction's Bridge A explicitly treats the Mn-toxin model and the α -synuclein-driven model as plausibly converging on a shared C1q/C3-CR3 downstream node despite different upstream triggers — an inference, not an observed fact, and now further weakened by the critic's inability to verify PMID 42531937 at all.
- **"Dual role" framing (Bridge B)**: reconstruction concluded C1q plays a genuinely dual role (pathological pruning trigger and protective clearance component) based on the dissociated PMID 41554750 outcome. The critic's stress-test found this **understates the risk**: the paper's own framing is not two independent "roles" but a single, temporally biphasic mechanism (transient protective clearance → macrophage exhaustion → accelerated pathology) — and a chronic, fully-neutralizing antibody like ANX005 is arguably a *closer* proxy to a constitutive knockout (which trended toward harm) than the one, unverified toxin-model paper is to a positive case.

5.6 Unresolved relationships

- Whether PMID 42531937's "neutralizing C1q function" was pharmacological (ANX005-like) or genetic — unconfirmed by either agent, a first-order gap given this is the sole positive functional-rescue citation.
- Whether ANX005's confirmed CSF saturation (HD trial) reflects meaningful nigrostriatal-synapse-level target engagement, as opposed to a bulk CSF-compartment measure — no region-specific data exist.
- Full-text verification of PMID 42477211, 42481480, and 42263400 — all inaccessible to the critic this validation, all originally reported only by the reconstruction agent.

5.7 Contradictory evidence

Two independent contradicting/complicating findings, both surviving critic scrutiny:

1. **PMID 41554750**: genetic C1q deletion worsens enteric histopathology in a PD-relevant α -synuclein model (male mice) — a direct hit on ANX005's own target, verified verbatim.
2. **New, critic-sourced**: ANX005's own human CNS trial (NCT04514367) shows CSF C4a and CSF NfL both *rising* during dosing — directionally unfavorable to the "functional target engagement" and "clinical activity" framing the reconstruction drew from the same trial's published paper.

5.8 Critic findings (summary — full report in agent transcript, agentId [a28b89f1d4d18e742](#))

The critic independently reconfirmed the ClinicalTrials.gov and Monarch findings, verified PMID 41554750/37961460 verbatim via live search (the only PMID it could fully verify this validation), and could not access five of the six most load-bearing PubMed citations despite extensive effort — explicitly flagged as a real gap, not treated as tacit support. It supplied genuinely new evidence (the CT.gov registry biomarker trends) not present in the reconstruction. Critic's own words: *"The verdict against ANX005... is not airtight either, and those PMIDs deserve re-verification once source access is restored. But per the governing instruction not to rescue the hypothesis absent that verification, and given the material new contradictory evidence found independently, UNDERMINED is the honest call."*

5.9 Main evidence gap

The single positive functional-rescue citation for C1q specifically (as opposed to downstream C3/C3aR) — PMID 42531937 — remains unverified by either agent across the entire pipeline. Every other paper that causally/functionally tests this pathway in a PD-relevant model implicates C3/C3aR, not C1q, or (in the one genuine C1q loss-of-function test available) shows a mixed, partly-unfavorable result.

5.10 Explicit conflicts/revisions relative to initial screening / N/P critic

- initial screening found "no PD- or C1q-specific PD literature" — this validation found several (PMID 42477211, 42481480, 42531937), a genuine upgrade, but the single functionally-causal one (42531937) could not be verified by either agent.
- initial screening flagged CNS/BBB penetration as the "major open question" — this validation's HD trial data (PMID 41787606) is more nuanced than a simple resolution: CSF saturation is confirmed, but the critic's independent registry pull shows rising CSF C4a/NfL, meaning saturation did not translate into confirmed functional pathway blockade or reduced neurodegeneration markers in that same trial.
- New contradicting evidence (PMID 41554750, and the CT.gov biomarker trends) did not exist in any form accessible during initial screening or the N/P critic.
- N3/P4 was retained after N/P critic reassessment and was independently re-examined during this validation.

5.11 Final verdict: **HYPOTHESIS_UNDERMINED**

5.12 Justification

The hypothesis requires that C1q itself, not merely the downstream complement cascade, be a net-beneficial target to block in PD — and the two most directly relevant pieces of functional evidence located this validation cut against that: the one genetic loss-of-function test in a PD-relevant model showed a mixed, partly-harmful outcome on C1q's own node, and ANX005's own human CNS trial shows rising CSF markers of complement activation and neurodegeneration during dosing, not falling ones. The sole paper that would offset this — direct functional rescue from C1q neutralization in a Parkinsonian model (PMID 42531937) — could not be verified by either agent despite repeated, good-faith attempts across the whole pipeline, so it cannot be relied upon to counterbalance the confirmed contradicting evidence. Per the explicit rule against rescuing a hypothesis absent verification, this is recorded as undermined, with the caveat that the single unverified citation, if it is confirmed accurate on re-access, would meaningfully change this picture and should be re-checked before this verdict is treated as final.

6. BIS-T-23 (syn. Ryngo 1-23)

6.1 Identity

Bis-tyrphostin-derivative research-tool compound, positive allosteric activator of DNM1/DNM2 (dynamin) self-assembly/GTPase activity. No CAS/CID resolved (UNRESOLVED, outside approved-source scope). **Bis-T-23 = Ryngo 1-23 identity claim downgraded to SUPPORTED_CONTEXT** (not OBSERVED_LITERATURE) by both agents — PMID 25498249 dual-indexes both names as MeSH substance terms and calls the compound "a bis-tyrphostin derivative, Ryngo® 1-23," but no single source explicitly states the synonymy. The critic independently confirmed this is the only co-occurring paper and that no accessible source states the equivalence outright — the reconstruction's caution was warranted, "if anything under-cautious."

6.2 Hypothesis (as carried into this validation)

Bis-T-23, by promoting dynamin-mediated membrane fission, could restore/enhance synaptic vesicle (SV) endocytosis and thereby counteract the endocytosis defects implicated in monogenic PD (DNAJC6/auxilin loss-of-function; α -synuclein-AP2 pathway dysfunction), protecting dopaminergic neurons.

6.3 Mechanism chain — REVISED from initial screening, corroborated and sharpened by the critic

initial screening flagged, as an open question, that the PD-linked endocytosis defect "maps more to auxilin-mediated uncoating, a downstream step" than to the fission step Bis-T-23 targets. This validation resolves the branch into three distinct, non-interchangeable steps of the canonical clathrin-mediated endocytosis cycle (confirmed verbatim via PMID 21779028): **(a)**

AP2-dependent cargo/adaptor recruitment (α -synuclein's role, PMID 40216248, 25009269 — upstream), **(b)** dynamin-mediated fission (Bis-T-23's actual target), **(c)** auxilin/Hsc70-mediated uncoating (the DNAJC6-PD-linked step, PMID 41935042, 36920906/Vidyadhara 2023 — confirmed verbatim by the critic, no fission defect reported in the auxilin-KO PD mouse model). **The PD monogenic genetics implicate step (c), not step (b).**

A partial complication (not a resolution) was raised by the reconstruction: older Sever-lab biochemistry (PMID 16946707, 12791276; non-neuronal, 2003–2006) shows dynamin:GTP and auxilin are physically/functionally coupled even at early, pre-fission steps — meaning the steps are not fully independent. **Neither agent could access these two papers' full text** to confirm the specific claim (auxilin depletion reduces coated-pit numbers) used to support this complication — flagged by both as a shared blind spot.

The critic's independent reasoning through this gap, conditional on the reconstruction's own stated Edge 9, points the "coupling" evidence in the opposite direction from a rescue: if auxilin loss creates a downstream bottleneck (fewer new pits form because the free clathrin/adaptor pool isn't recycled), then artificially forcing *more* fission events via a dynamin activator onto that bottlenecked system would be expected to deplete the pool faster, producing a **futile-cycle risk** rather than compensation. This is explicitly flagged by the critic as **AGENT_INFERENCE** on an unverified primary source, not confirmed fact — but it sharpens rather than resolves the reconstruction's neutral framing, and does not support treating the coupling evidence as a mitigating factor.

A genuinely new but ultimately non-transferable data point was found: **PMID 25939402** shows Ryngo 1-23 affects the *exocytotic* fusion pore (increases catecholamine release per vesicle) in chromaffin cells — the first neurosecretory-cell data located for this compound (narrowing, but not overturning, screening-stage "zero CNS/BBB data of any kind" to "zero data on the endocytic-fission-relevant phenotype in any neuronal/neurosecretory cell type"). Both agents agree this concerns a mechanistically distinct dynamin function (exocytosis, not endocytosis) and does not transfer to the PD hypothesis; the critic additionally noted the paper's own title calls dynamin a "bi-directional regulator," meaning the specific direction of effect (increase vs. decrease) could not be confirmed by either agent from the retrievable text.

6.4 Edge-level evidence ledger (key edges)

#	Subject → Object	Status	Source	Statement
1	Bis-T-23 → DNM1/DNM2 oligomerization/ GTPase activity	OBSERVED_LITERATURE, confirmed & strengthened by critic	PMID 25962121, 27432739, 30350425, 41077241	All four independently confirmed accurate; critic's full "Bis-T-23" search (15/15 titles) found completely non-neuronal literature — kidney/podocyte/ renal-fibrosis (11), cancer-migration (1), adjacent renal topics — stronger than the reconstruction's "almost entirely"
2	Canonical CME order: AP2 cargo selection → coat building → dynamin fission → auxilin uncoating	OBSERVED_LITERATURE, verified verbatim by critic	PMID 21779028	Confirms fission and uncoating are sequential, temporally distinct steps
3	Auxilin-KO mouse → PD-like phenotype (uncoating/SV-sorting defect, no fission defect reported)	OBSERVED_LITERATURE, verified verbatim by critic	PMID 36920906 (Vidyadhara 2023)	Title and abstract opening independently confirmed: "Dopamine transporter and synaptic vesicle sorting defects

#	Subject → Object	Status	Source	Statement
				underlie auxilin-associated Parkinson's disease"
4	DNAJC6 (auxilin) biallelic LOF → juvenile-onset PD (monogenic)	OBSERVED_LITERATURE + OBSERVED_KG, independently re-derived from scratch by critic	PMID 33983693, 26528954; Monarch MONDO:001423 1	Critic queried MONDO:001423 1 directly and confirmed DNAJC6 (alias PARK19) is the sole causal gene on that disease record — DNM1/DNM2 do not appear
5	DNM1, DNM2 → PD/parkinsonism (gene-disease)	UNRESOLVED (zero KG association), independently reproduced by critic from scratch	Monarch	DNM1: epileptic encephalopathy only. DNM2: myopathy/CMT/fetal-akinesia only. Zero PD entries for either — Bis-T-23's actual pharmacological targets have no curated PD linkage at all.
6	LRRK2-phosphorylated auxilin → SV endocytosis defect in patient-derived DA neurons; rescued by restoring auxilin	OBSERVED_LITERATURE, verified verbatim by critic	PMID 29735704	"Pathogenic phenotypes were partially attenuated by restoring auxilin function" — no dynamin-directed rescue tested;

#	Subject → Object	Status	Source	Statement
	(NOT by dynamin activation)			closest existing head-to-head PD-neuron test, and it favors the uncoating/auxilin axis, not fission
7	α -synuclein → binds/stabilizes AP2 on synaptic membranes (initiation step, upstream of dynamin)	OBSERVED_LI TERATURE (reconstruction; SOURCE_ACC ESS_FAILED for critic)	PMID 40216248	Not independently re-verified this validation
8	Ryngo 1-23 → exocytotic fusion-pore modulation (chromaffin cells)	OBSERVED_LI TERATURE (title only, confirmed by both agents); direction unconfirmed by critic	PMID 25939402	Only neurosecretory data point found; concerns exocytosis, not endocytic fission
9	Dynamin:GTP ↔ auxilin coupling at pre-fission steps	SOURCE_ACC ESS_FAILED for BOTH agents	PMID 16946707, 12791276	Load-bearing for the reconstruction's Bridge A; neither agent could verify from primary text
10	Bis-T-23 / Ryngo 1-23 → PD (novelty)	UNRESOLVED (zero-result, independently reproduced)	PubMed	"Bis-T-23 AND Parkinson" = 0 (both agents); "Ryngo 1-23" unqualified = 9 hits, independently reproduced exact count and

#	Subject → Object	Status	Source	Statement
				content by critic — gut L-cell secretion, CCT chaperonin screen, Chlamydia entry (×2), HIV, cancer migration, fusion-pore biophysics, vascular tone, plus the chromaffin paper — none neurodegenera tion/PD-relevan t ; critic confirms the original search was exhaustive, not incomplete
11	Bis-T-23 / Ryngo 1-23 → clinical trials	UNRESOLVED (zero)	CT.gov /api/int/studies	0 trials, any indication — expected for an early-stage research tool compound

6.5 Inferred bridges + reasoning (AGENT_INFERENCE)

- **Bridge A (dynamin/auxilin coupling as a possible rescue mechanism)**: unresolved at the source level (both agents' access to PMID 16946707/12791276 failed) and, per the critic's conditional reasoning from the reconstruction's own Edge 9, more plausibly points toward a **futile-cycle risk** than a compensatory mechanism if auxilin loss is a downstream-recycling bottleneck rather than a simple pile-up of fission-incompetent pits. Neither direction is confirmed; the critic's own assessment: *"this reasoning is AGENT_INFERENCE... it should not be read as confirming the futile-cycle hypothesis, only as showing that the coupling evidence... points toward 'compounds the problem' at least as plausibly as toward 'compensates for it.'"*

- **Branch-level plausibility vs. compound-specific relevance:** PD-linked genes acting at multiple distinct steps of the SV endocytic cycle (auxilin at uncoating, α -synuclein/AP2 at cargo recruitment, LRRK2 acting on auxilin) collectively support "SV endocytosis is a real, druggable PD mechanism" at the branch level — this does not transfer target-class credit to Bis-T-23 specifically, since its target step (fission) is not the one implicated by the causal genetics.

6.6 Unresolved relationships

- Whether pharmacological or genetic dynamin activation/fission-promotion rescues any auxilin/DNAJC6-deficiency phenotype, in any cell type — no paper tests this directly (confirmed absence by both agents' independent searches).
- Whether Bis-T-23/Ryngo 1-23 has any effect on synaptic vesicle *endocytosis* (as opposed to the exocytotic fusion-pore effect already found) at any real CNS or neurosecretory synapse.
- Whether DNAJC6/auxilin loss impairs the fission step itself (not just uncoating) in neurons specifically — the clearest relevant nuance (Sever-lab coupling data) is from 2003–2008, non-neuronal cell lines, and unverified by either agent this validation.
- Bis-T-23 = Ryngo 1-23 exact chemical identity — unresolved at the level of an explicit stated synonymy in any retrieved source.

6.7 Contradictory evidence

No directly **CONTRADICTED** edge was found by either agent (i.e., no source flatly negates a claimed relationship). What both agents independently converge on, and what functions as the practical equivalent of contradiction for this hypothesis, is a **structural absence of support at the compound's own target**: Bis-T-23's actual pharmacological targets (DNM1, DNM2) carry zero PD/parkinsonism disease association in Monarch — independently derived from scratch by both agents, including a direct query of the PD-linked disease record (MONDO:0014231) itself, which lists DNAJC6 as its sole causal gene. The canonical step-ordering (PMID 21779028) and the direct auxilin-KO PD mouse model (PMID 36920906, no fission defect reported) both reinforce that the causal PD lesion sits downstream of the step Bis-T-23 targets.

6.8 Critic findings (summary — full report in agent transcript, agentId **a7a40fcbed5618ceb**)

The critic independently verified essentially every checkable claim in the reconstruction (PMID 21779028, 36920906, 29735704 all confirmed verbatim; the Bis-T-23/Ryngo 1-23 literature searches exactly reproduced in count and content; the Monarch KG topology independently re-derived from scratch rather than trusted). It found the reconstruction's caution on the Bis-T-23/Ryngo-1-23 identity question was, if anything, under-stated, and it added a novel, unfavorable inferential reading of the one complicating factor (Bridge A). **Note on the critic's**

own verdict language: the critic titled its overall assessment "HOLDS UP," but its own text clarifies this means *the reconstruction's skeptical, target-mismatch conclusion holds up under independent stress-testing* — not that the therapeutic hypothesis (Bis-T-23 as a viable PD repurposing candidate) holds up. Read correctly against the task's own verdict vocabulary (which describes the therapeutic hypothesis, not the critique of it), the critic's substantive findings all point toward the therapeutic hypothesis being undermined, and this document's final verdict below reflects that substance rather than the critic's label. Critic's own words: *"My one genuine addition is on Bridge A: reasoning conditionally from the reconstruction's own cited Edge 9... a forced dynamin activator layered onto an auxilin-deficient, uncoating-bottlenecked system is at least as plausibly a futile-cycle risk as a compensatory mechanism."*

6.9 Main evidence gap

No study, in any cell type, has ever tested whether pharmacological dynamin activation affects an auxilin/DNAJC6-deficiency phenotype in either direction. Bis-T-23's entire pharmacological evidence base is non-neuronal (kidney/podocyte physiology, cancer-cell migration, one immune B-cell rescue, one chromaffin-cell exocytosis study) — there is no neuronal endocytosis data for this compound at all.

6.10 Explicit conflicts/revisions relative to initial screening / N/P critic

- the initial screening entry already flagged the fission-vs-uncoating step mismatch as an open question — this Deep Validation corroborates and sharpens that concern rather than resolving it in either direction, and adds the new structural finding that Bis-T-23's molecular targets (DNM1/DNM2) have zero PD-disease-association support in Monarch, a check not performed during initial screening.
- screening-stage F2 claim ("real activator phenotype... in kidney/podocyte disease") is confirmed accurate and, per the critic's exhaustive title-level re-search, is narrower than "almost entirely non-neuronal" — it is completely non-neuronal for the endocytic-fission mechanism, with one exocytosis-only exception (PMID 25939402) newly found this validation.
- screening-stage "zero CNS/BBB data of any kind" claim is revised to "zero data on the endocytic/fission-relevant phenotype in any neuronal or neurosecretory cell type" given the newly found chromaffin-cell exocytosis paper.
- N3/P4 was retained after N/P critic reassessment and was independently re-examined during this validation.

6.11 Final verdict: **HYPOTHESIS_UNDERMINED**

6.12 Justification

The mechanistic premise requires that promoting dynamin-mediated membrane fission address the PD-relevant endocytosis defect — but every source of causal PD genetics located by either agent (DNAJC6/auxilin monogenic PD, LRRK2-phosphorylated-auxilin sporadic PD) implicates the uncoating step, not fission, and the one paper offering a head-to-head rescue test in patient-derived dopaminergic neurons succeeded via restoring auxilin, not via any dynamin-directed intervention. Independently, from scratch, both agents confirmed Bis-T-23's actual molecular targets (DNM1, DNM2) carry zero curated Parkinson's-disease association, while the sole PD-linked gene in this entire pathway (DNAJC6/auxilin) is a dynamin-*interacting* protein, not dynamin itself. The one complicating factor that could soften this picture (dynamin/auxilin biochemical coupling) rests on two papers neither agent could access, and the critic's conditional reasoning through the reconstruction's own summarized evidence suggests this coupling, if anything, points toward a futile-cycle risk rather than a rescue mechanism. Combined with a completely non-neuronal primary pharmacology evidence base for the endocytic mechanism in question, this is recorded as undermined — a target/step mismatch reinforced by independent, from-scratch verification on both the mechanistic-reconstruction and critic passes, not merely an unproven hypothesis.

Summary

Candidate	Screening decision	This Deep Validation's final verdict	Central reason
Oltipraz	KEEP	INCONCLUSIVE	No study has ever tested oltipraz in any PD-relevant model (cellular, animal, or human), and the one human trial that measured whether oral oltipraz reliably engages its own proposed Nrf2 target found that it does not

Candidate	Screening decision	This Deep Validation's final verdict	Central reason
PMX205	KEEP	HYPOTHESIS_UNDERMINED	Receptor-level mismatch, complement-silent PD-model pruning evidence, and a directly contradicting genetic finding
Neflamapimod (VX-745)	KEEP	INCONCLUSIVE	Zero direct PD/dopaminergic evidence in either direction; real DLB mechanism with an unclosed translation gap
Emricasan (IDN-6556)	KEEP	HYPOTHESIS_UNDERMINED	Genetic proxy test (Casp3-KO, MPTP model) contradicts the apoptosis-inhibition logic; zero KG support at every mechanism node
ANX005 (Tanruprubart)	KEEP	HYPOTHESIS_UNDERMINED	Genetic C1q loss-of-function shows mixed/harmful histopathology result; ANX005's own human CNS trial shows rising, not falling, CSF complement/neurodegeneration markers; sole positive functional citation unverifiable

Candidate	Screening decision	This Deep Validation's final verdict	Central reason
Bis-T-23 (Ryngo 1-23)	KEEP	HYPOTHESIS_UNDERMINED	PD-causal genetics implicate the uncoating step, not the fission step Bis-T-23 targets; its actual molecular targets have zero KG-level PD association

All six Deep-Validated retained candidates have either been undermined or remain unproven under independent, adversarial, edge-level scrutiny — none has emerged with a verdict of HYPOTHESIS_HOLDS_UP. Ru265 and Ruxolitinib remained outside the Deep Validation scope after both were reclassified to REJECT during the N/P critic following newly identified PD-related exposure.

No therapeutic efficacy claims are made anywhere in this document. All verdicts describe the strength of the *mechanistic hypothesis* as tested against the three approved evidence sources, not clinical recommendations.